ADAPTIVE SOFTWARE ENGINEERING (ASE) COURSE PROJECT SUBMISSION

PROJECT TITLE:

Context-Aware Self-Healing Distributed Systems Using Generative Graph Neural Networks

for Dynamic Fault Remediation

TEAM DETAILS & ROLE ALLOCATION:

Member 1 (AI Architect):

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REPOSITORY & ASSET DETAILS:

• GitHub Repository: https://github.com/Adityachintada/ASE-PROJECT

• Branch: main

• Submission Date: August 10, 2026

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PROJECT ABSTRACT

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TITLE:

Context-Aware Self-Healing Distributed Systems Using Generative Graph Neural Networks for Dynamic Fault Remediation

ABSTRACT:

Modern distributed architectures face rapid cascading failures that static rule-based monitoring cannot efficiently isolate. This project introduces an autonomous self-healing framework driven by Generative Graph Neural Networks (GNNs) integrated into a MAPE-K control loop. Telemetry and service dependencies are modelled as dynamic spatio-temporal graphs. A Generative Graph Autoencoder detects structural anomalies, reconstructs baseline topologies, and synthesizes context-aware remediation workflows—including automated traffic rerouting and dynamic node isolation. Evaluated under synthetic chaos benchmarks, the system significantly reduces Mean Time to Remediation (MTTR) and prevents cascading system downtime.

KEYWORDS:

Self-Healing Systems, Generative GNNs, MAPE-K Loop, Fault Remediation, Distributed Systems.